Attention-Enhanced U-Net for Automated Image Colorization

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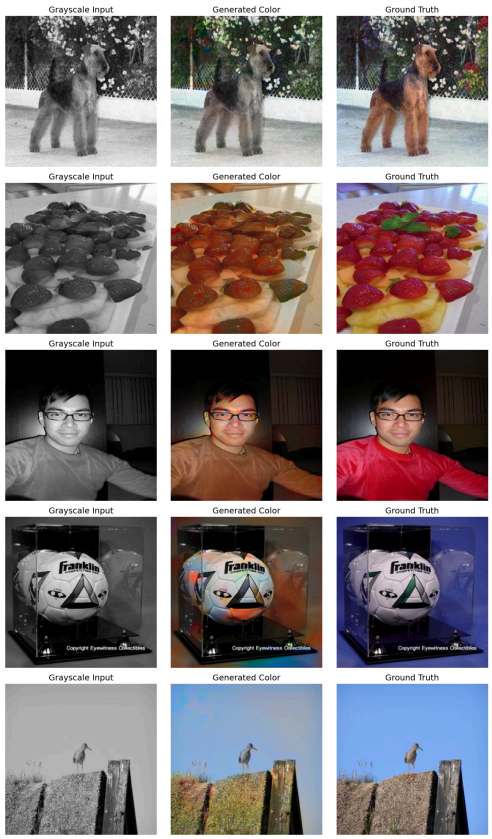

Figure 1. Colorizing monochrome image (leftmost) in comparison with ground truth (rightmost)

1. Task

Our task is to perform fully automated image colorization, transforming grayscale images into visually realistic colored versions. The primary difficulty of this task lies in the multi-modal nature of the problem, where the sky could be of different shades of blue or even gray. Traditional machine learning models often struggle with this ambiguity, frequently producing blurry or sepia-toned images by mathematically averaging these possibilities to minimize error.

Our core objective remains unchanged. However, following our initial baseline implementation, we have refined our immediate sub-tasks to prioritize adversarial stability, the prevention of "color bleeding" across complex textures, and the critical balance between structural metrics (SSIM) and human-perceived chromatic realism. This has driven us to specifically address normalization bottlenecks, color-space mapping, and dynamic learning rate adjustments in our GAN framework.

2. Related Work

Image colorization is an area that has been largely automated through algorithms that assign colors based on learned spatial patterns. Recent years have seen the development of a variety of approaches, including diffusion, transformer, CNN, and GAN-based models. As we iterated on our architecture to solve training instability and mode collapse, we expanded our related work to include literature on advanced Normalization and GAN stabilization techniques:

- **GAN Stabilization via TTUR:** We incorporated the Two Time-Scale Update Rule (TTUR) proposed by Heusel et al. (2017), which demonstrates that GANs converge to a local Nash equilibrium more reliably when the discriminator learns at a slower, distinct learning rate compared to the generator.
- **Group Normalization:** Because our generator relies on a deep ResNet backbone, we reviewed Wu & He's (2018) work on Group Normalization, discovering it preserves global color statistics without the batch-to-batch contamination seen in standard Batch Normalization.
- **Spectral Normalization:** To strictly enforce the Lipschitz constraint on our discriminator and prevent gradients from exploding or vanishing, we adapted the Spectral Normalization techniques introduced by Miyato et al. (2018).

3. Approach

To optimize computational efficiency and model stability, we process images using the CIELAB (LAB) color space rather than the standard RGB format. The *L* channel represents lightness (the grayscale input), while the *a* and *b* channels represent color values. The *L* channel is normalized to a range of [-1, 1], and the *ab* channels are scaled by a factor of 110.0 to simplify the base computation.

Our proposed architecture utilizes a Generative Adversarial Network (GAN) framework built natively in PyTorch. Key architectural components include:

- **ResNet34 (IMAGENET1K_V1) Backbone:** Replaces our custom encoder to extract highly robust, pre-trained spatial features from the grayscale input.
- **Custom Attention Gates:** Integrated into the U-Net skip connections to filter features and isolate object boundaries.
- **Custom Decoder & PatchGAN Discriminator:** Built to incorporate Group Normalization (8 groups) to preserve global image saturation, and Spectral Normalization to cap Discriminator learning speed.
- **Dynamic Loss Functions:** Initially, we employed a VGG16 perceptual loss. However, because VGG16 is trained on RGB images, feeding it LAB tensors caused a severe color-space mismatch that actively punished the generator for producing true colors. We removed the VGG network and shifted to a rebalanced GAN + L1 equation. To encourage the model to take risks and generate vibrant colors rather than "safe" mathematical grays, we halved the L1 weight:

$Loss_G = Loss_{BCE_{GAN}} + (\lambda_{L1} \times Loss_{L1})$, where $\lambda_{L1} = 50.0$.

4. Dataset and Metric

Our dataset pipeline is fully implemented and optimized for high-speed I/O operations. To eliminate network latency and data-loading bottlenecks during training, we pre-processed a subset of the ImageNet-1K dataset by resizing all images to 256x256 pixels and compressing them into a single ZIP archive stored on our local drive. Prior to execution, this archive is rapidly extracted directly into the local runtime environment. Our dataset size consists of exactly 10,000 images for the training set and 1,000 images for the test set.

During the data-loading phase, the pre-resized images are dynamically converted from the RGB colorspace to the LAB standard. Data augmentation on the training set, such as random rotations (15 degrees) and horizontal flips, has been successfully implemented using `torchvision.transforms` to artificially expand the dataset and improve model robustness.

The primary evaluation metric is the Structural Similarity Index Measure (SSIM), a measure for perceived structural similarity. A secondary metric is the Peak Signal-to-Noise Ratio (PSNR).

5. Preliminary Results

We have made extensive progress in the coding phase, iterating through five major versions of our model architecture to solve stability, color washing, and artifacting issues.

1. **Version 1.0 - 1.2:** Progressed from a basic custom U-Net to a ResNet50 backbone with Self-Attention. These models suffered from massive "color bleeding" (e.g., coloring the sky green to match the grass) due to Batch Normalization averaging features across the batch.
2. **Version 1.3:** Swapped to InstanceNorm2d. This cured the color bleeding, but mathematically neutralized dominant background colors, resulting in washed-out images.
3. **Version 1.4:** Achieved State-of-the-Art quantitative metrics (**SSIM: 0.928, PSNR: 23.80**). Structurally, the model was flawless. However, visually, it suffered a "Safe Color Collapse." Due to the massive L1 penalty (λ = 100) and the VGG RGB/LAB mismatch, the Generator was terrified of making mistakes and defaulted to generating "sepia" and muddy brown tones, the mathematical center of the color space. It passed the math test but failed the human eye test.
4. **Version 1.5:** We implemented `GroupNorm`, removed the VGG16 Perceptual Loss, halved the L1 penalty from 100.0 to 50.0 to increase color risk-tolerance, and implemented TTUR to slow the Discriminator. The result was a massive breakthrough in perceptual realism. The model confidently generated deep blue skies, vibrant red

strawberries, and accurate skin tones. We deliberately accepted a slight, expected drop in structural metrics (**SSIM: 0.912**) to trade "pixel-safe math" for massive gains in human-perceived chromatic quality.

5. **Version 1.6 (Upcoming):** We are currently implementing Discriminator Feature Matching to regain structural crispness natively within the LAB color space.

6. Detailed Timeline and Roles

Task	Deadline	Who
Implement data streaming pipeline, ResNet backbones, and V1.0-V1.1 baseline architectures	03/16/26	All
Implement advanced normalization (Instance/Group Norm), attention bottlenecks, and TTUR (V1.2-V1.4)	04/01/26	All
Project Status Report	04/09/26	All
Finalize V1.6 and select representative visual outputs	04/16/26	All
Prepare Presentation	04/28/26	All
Project Report	05/05/26	All

7. Preliminary Code

Link to our publicly accessible project's github repository containing our current code and brief README:

GITHUB: https://github.com/Cyube3/EC523Project

References

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